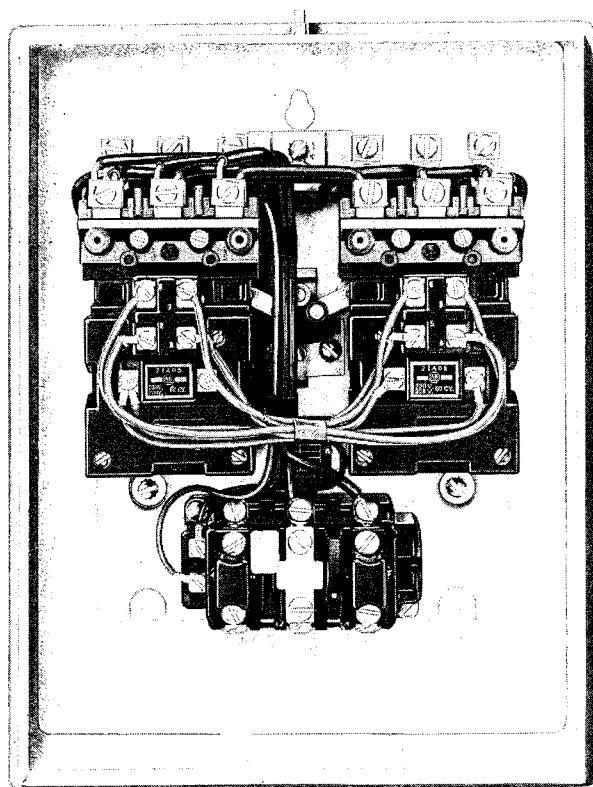




## Bulletin 705

## The Bulletin 705 Full Voltage Reversing Starter



**Bulletin 705, Size 1, Style RT, Nema 1  
with Cover Removed**

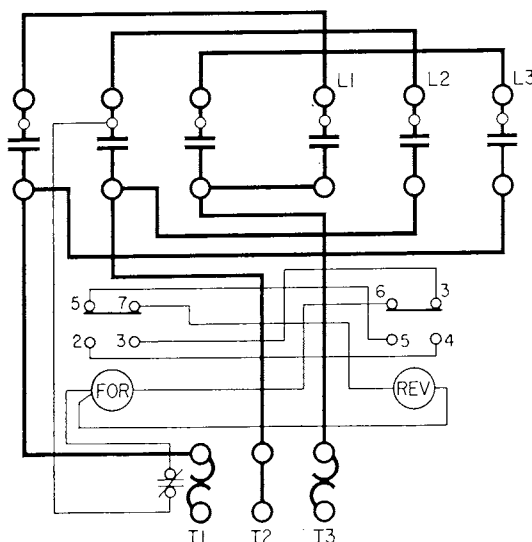
**THE BULLETIN 705 AS  
REPRESENTED BY A WIRING  
DIAGRAM**

A photograph and a wiring diagram of an Allen-Bradley Bulletin 705 reversing starter are shown on this page. The two can be compared as was done with the Bulletin 709. This should help make the meaning of the wiring diagram clearer and should be quite useful when making connections to an actual starter.

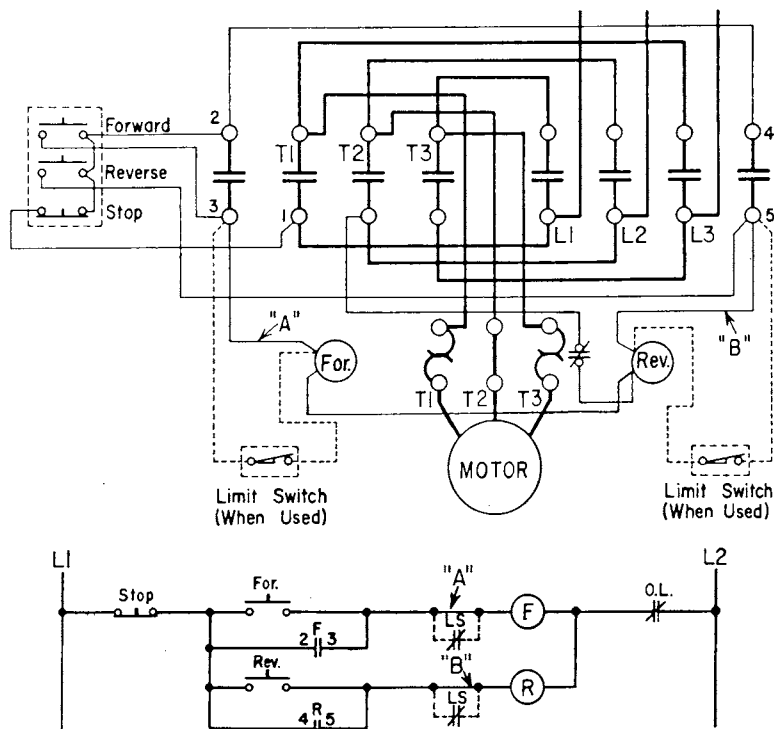
Because of the more involved wiring of the Bulletin 705, all of the connections cannot be clearly seen in the photograph. However, comparison with the wiring diagram should assist in tracing any wire on the photograph. Only the wires most likely to be worked with on the standard starter have been labeled.

Each reversing starter is equipped with a mechanical interlock. However, the mechanical interlock, armature, crossbar and overload reset mechanism are not shown in the wiring diagram since these parts need not be considered from a wiring standpoint.

**Wiring Diagram of  
Bulletin 705, Size 1, Style RT  
with Block Overload Relays**



### 3 Phase Starters

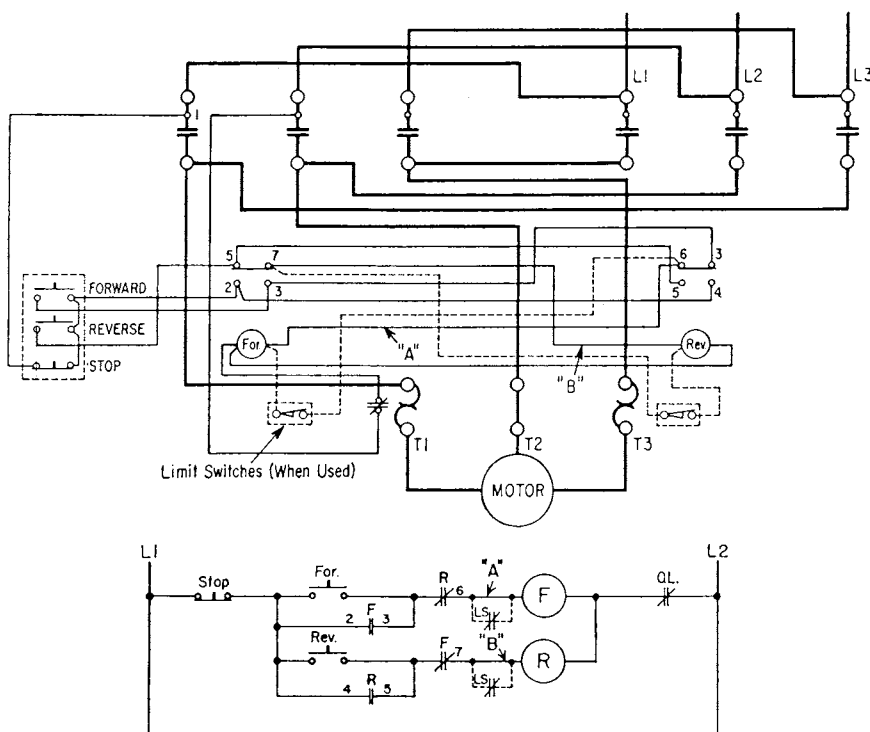
**Size 00**

Standard wiring with "Forward-Reverse-Stop" push button station. The "STOP" button must be depressed before changing directions.

A mechanical interlock is provided, however electrical interlocks are not furnished on size 00 reversing starters.

Limit switches can be added to stop the motor at a certain point in either direction. Connections "A" and "B" must be removed when limit switches are used.

**41.**



### Sizes 0, 1, 2, 3 and 4

Standard wiring with "Forward-Reverse-Stop" push button station. The "STOP" button must be depressed before changing directions.

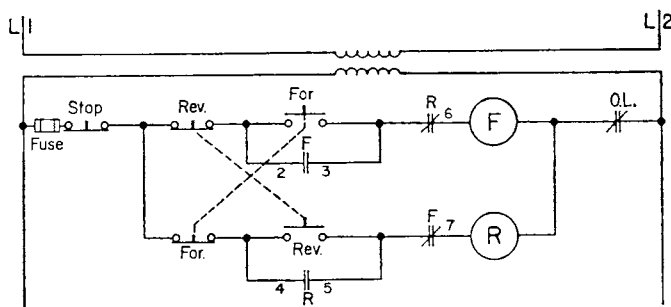
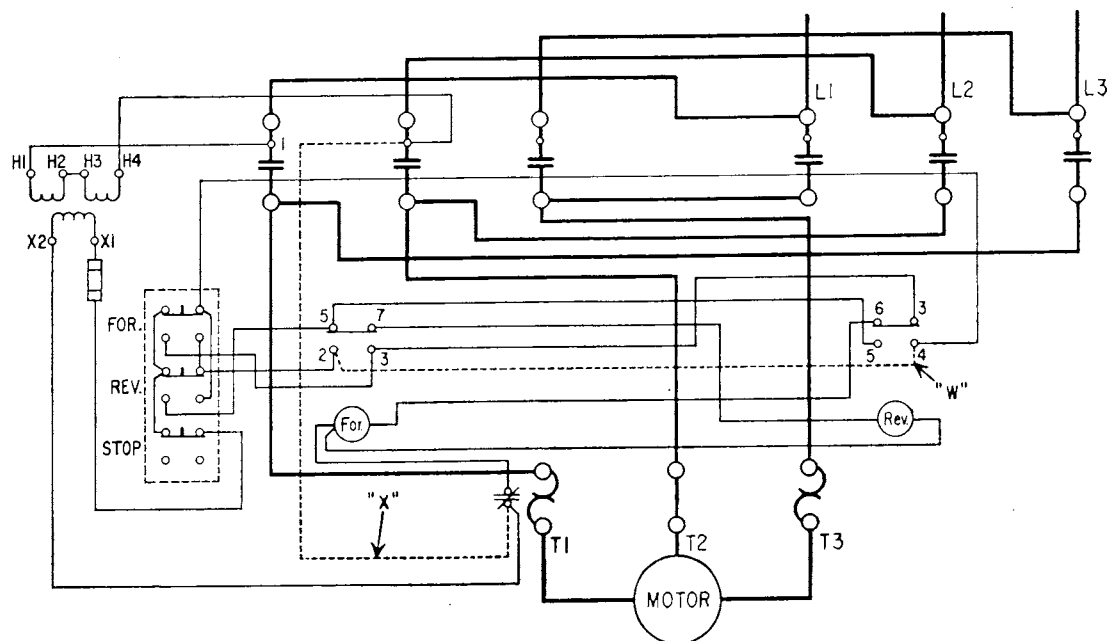
A mechanical interlock and electrical interlocks are supplied as standard on all reversing starters size 0 and larger.

Limit switches can be added to stop the motor at a certain point in either direction. Connections "A" and "B" must be removed when limit switches are used.

**42.**



## Push Button Station Variations

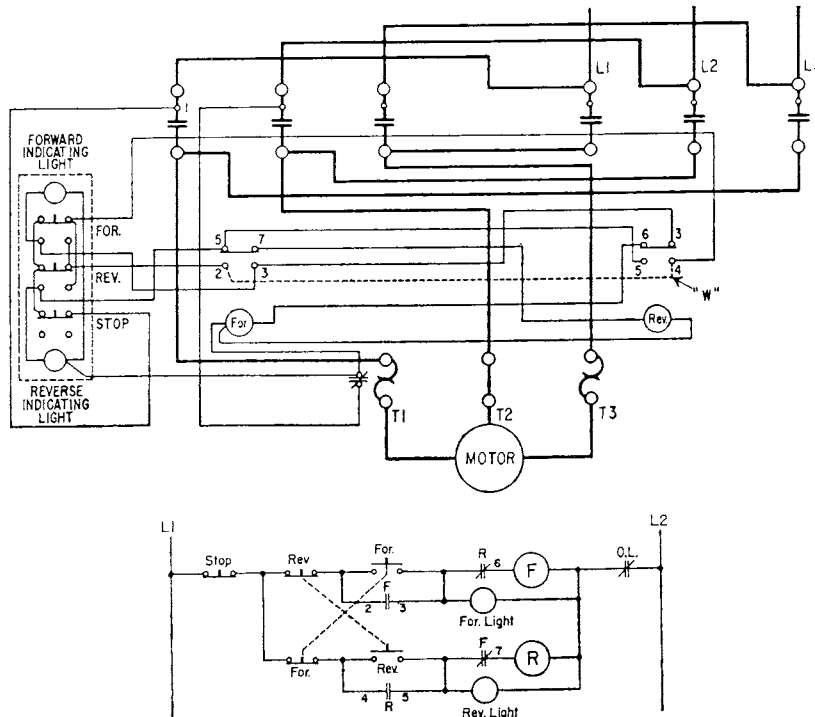


### Push Button Wired so Starter Can Be Switched from One Direction to the Other Without Pushing Stop Button

This scheme allows immediate reversal of the motor when it is running in either direction. It is not necessary to depress the "Stop" button when changing direction. A standard Bulletin 705 reversing switch can be used if wire "W" is removed.

The diagram shows the control circuit set up for reduced voltage control, although this may not be necessary in many cases. Notice that wire "X" must be removed when reduced voltage control is used. The push button station can be a Bulletin 800, Type 3SA (standard duty), Type 3HA (heavy duty) or a Bulletin 800T, Type 3TA (oiltight).

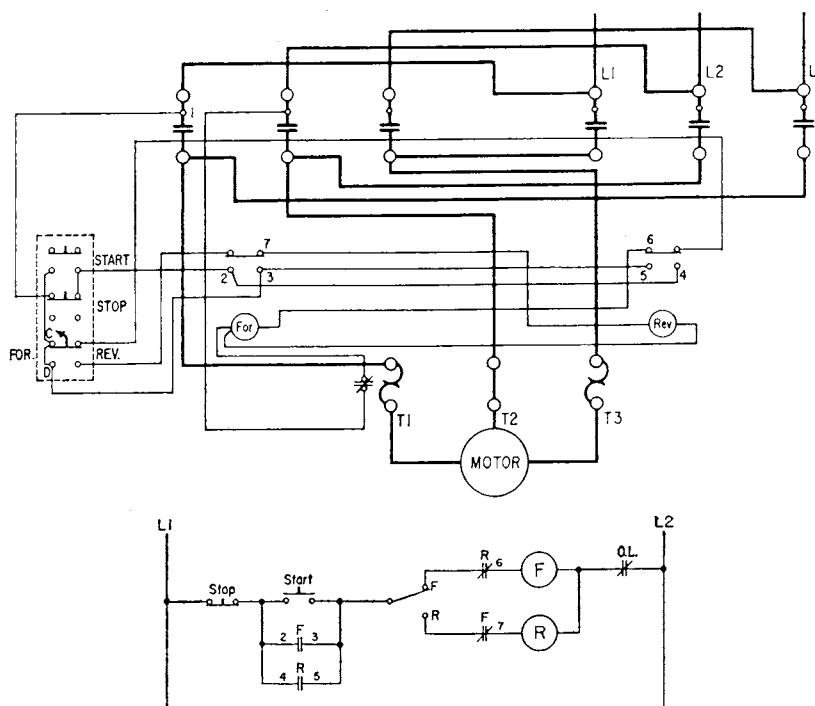
## Push Button Station Variations



### Starting and Stopping in Both Directions. Lights Indicate Direction in Which Motor is Operating

This setup provides exactly the same operation as shown in the previous diagram, except that pilot lights have been added to show which way the motor is running. Once again, standard Bulletin 705 reversing switch can be used if wire "W" is removed. The pilot lights can either be separately mounted or mounted in the push button station. If they are to be mounted in the station a Bulletin 800, Type 3HA2P can be used.

44.



### The Motor Runs in a Pre-selected Direction Which is Determined by The Setting of a Selector Switch

The motor can be run in either direction, but the desired direction must be set on a selector switch before starting. The motor is then operated from a "Start-Stop" station as a single direction motor.

The wiring of the standard Bulletin 705 reversing starter must be modified slightly to fill this requirement. Note that the connections which normally lead from the electrical interlock contacts to points 3 and 5 have been removed and that different connections have been made to the electrical interlocks and points 3 and 5.

It is usually most convenient to include the selector switch as part of the push button station. This can be done with either a Bulletin 800 heavy duty station or a Bulletin 800T oiltight station.

45.

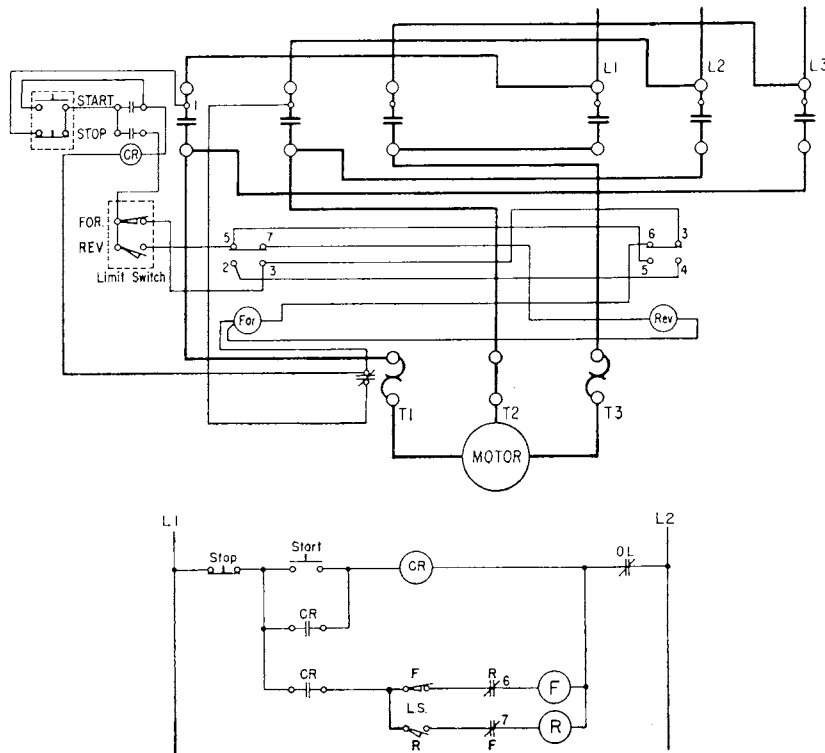


### Push Button Station Variations

#### Limit Switch Controls Reversing

Here the direction of the motor is determined by the position of a limit switch. A start-stop push button station is used to energize the system and the motor will start according to the position of the limit switch. The wiring of the standard Bulletin 705 need not be modified for this type of operation. Limit switch connections are made directly to the electrical interlocks.

It is necessary to use a control relay in this system and this relay is a Bulletin 700, Type C-20. The limit switch can be any of several in the Bulletin 802T line having one NO and one NC contact.



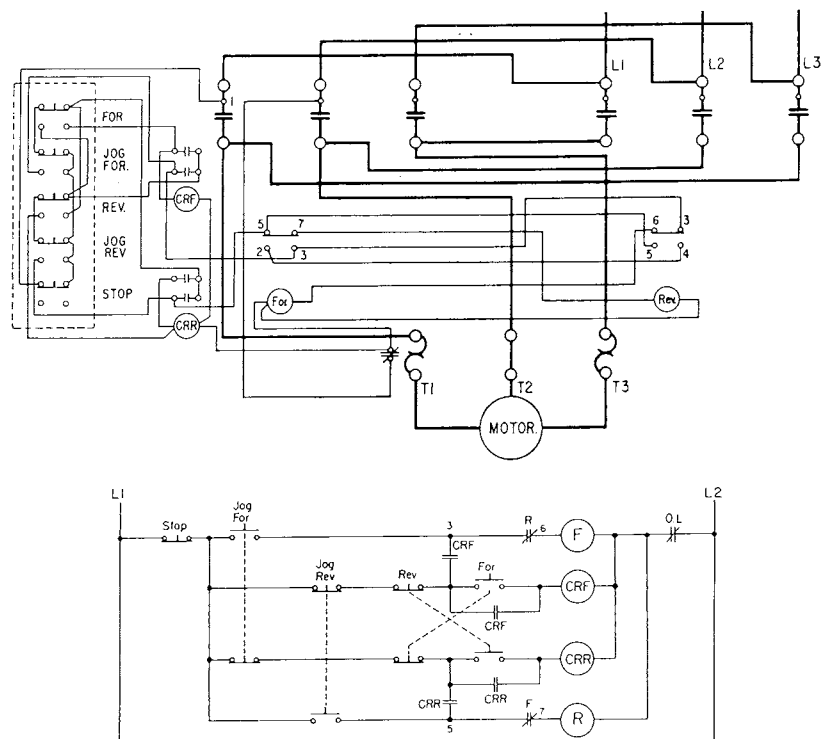
46.

### Jogging

#### Jogging With Relays

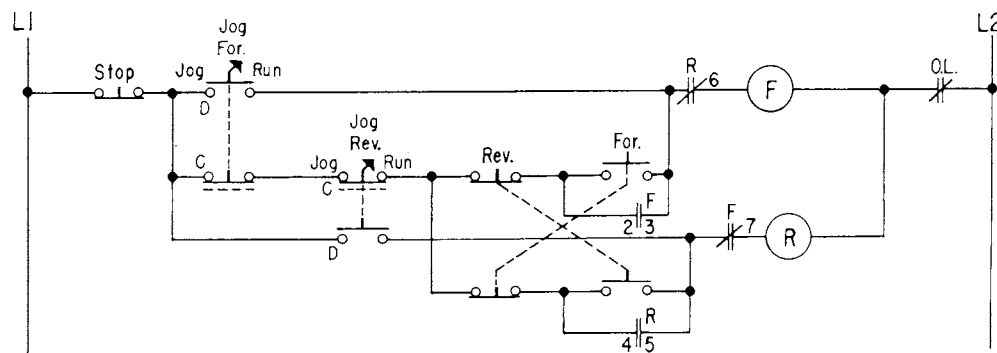
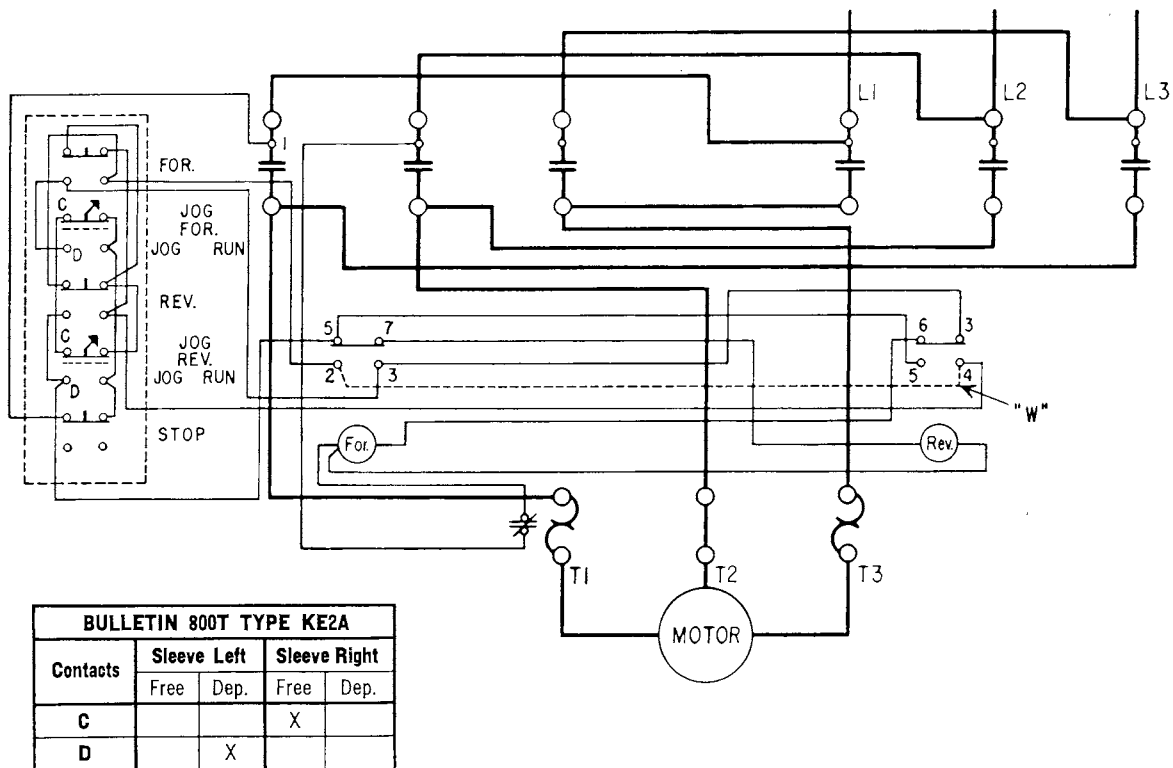
In this arrangement for jogging and running in either direction "jogging relays" are used to insure safe jogging. These relays prevent either the forward or reverse contactor from locking in during jogging.

The push button station can be a Bulletin 800, heavy duty, or a Bulletin 800T oiltight. Bulletin 700, Type C-20 relays may be used.



47.

## Jogging



## Starting, Stopping and Jogging in Either Direction

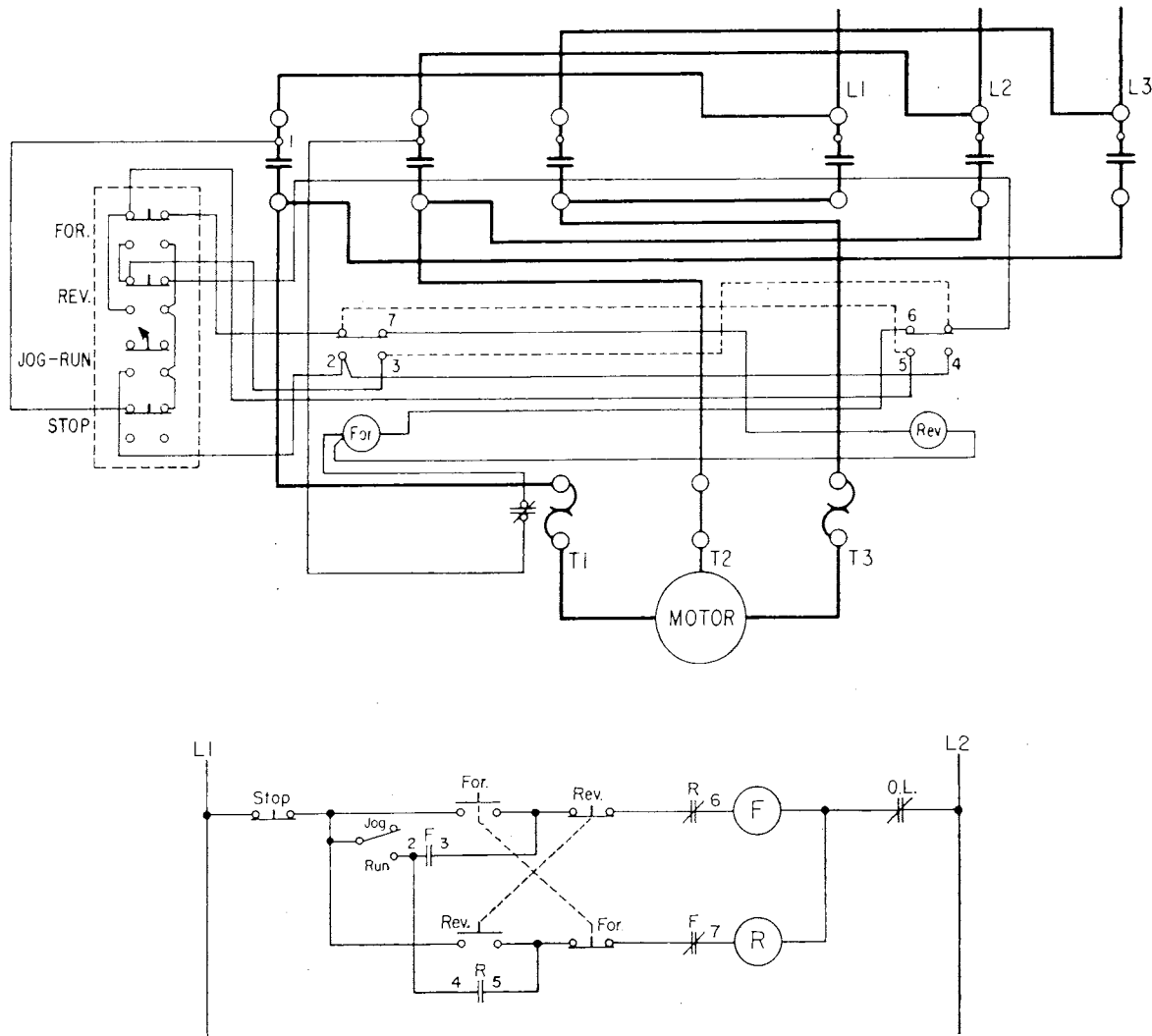
This is one of the safest and most desirable methods of jogging a reversing starter. The motor will run steadily in either direction and also can be jogged in either direction.

Bulletin 800T Type KE2A jogging units in an oiltight enclosure are shown in the diagram. If desired, a Bulletin 800H, heavy duty, push button station could also be utilized.

To jog, the jogging ring must be turned to the "JOG" position. The NC contact of the jogging unit is held open thus insuring safe action by preventing either contactor from locking in during jogging.



## Jogging



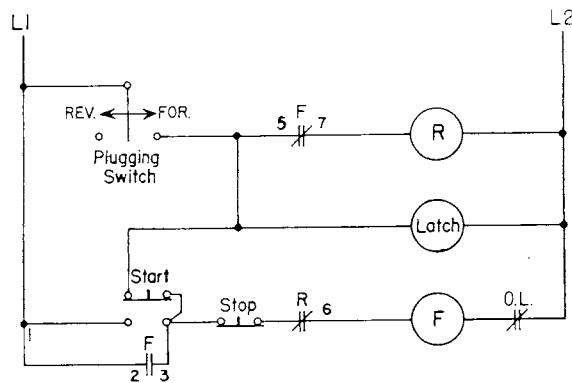
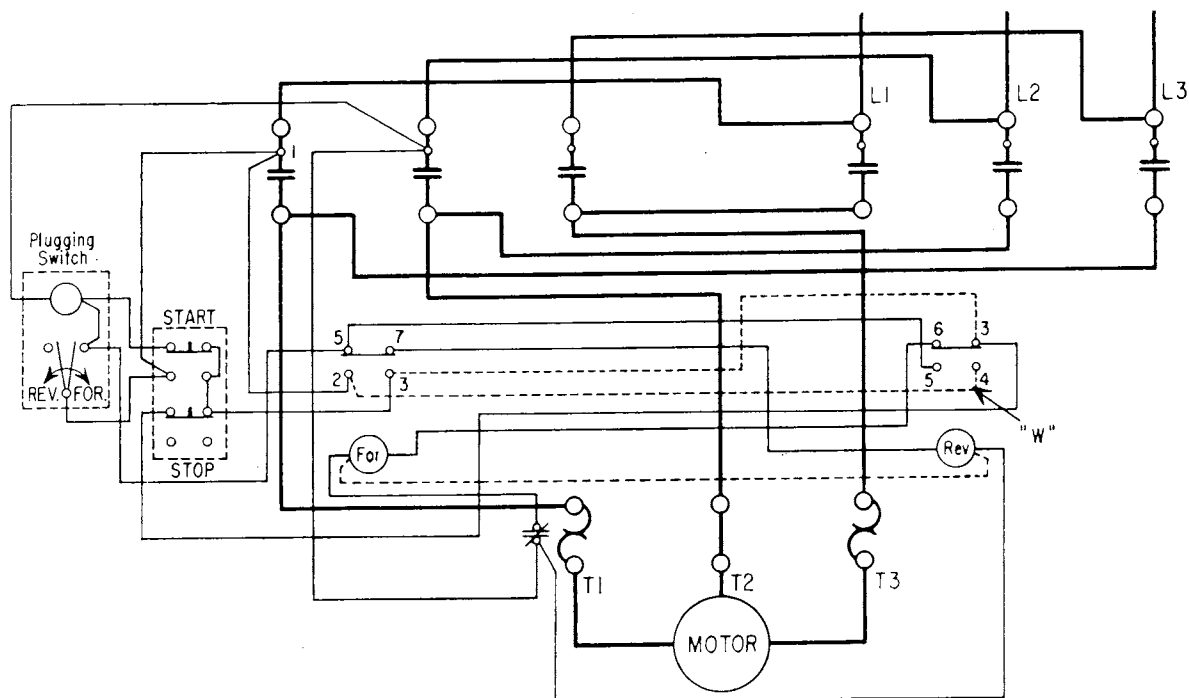
### Starting, Stopping and Jogging in Either Direction. Jogging Controlled Through a Jogging Selector Switch

Here, the motor can run normally in either direction or can be jogged in either direction. With the selector in the "run" position, the motor can be started in either direction and will stop when the stop button is pressed. It is not necessary to press the stop button before changing from forward to reverse.

With the selector in the "jog" position, the "forward" and "reverse" buttons act as jogging buttons. The motor will run in the indicated direction when one of them is pressed but will stop as soon as the button is released.

The wiring of the standard Bulletin 705 must be modified for this type of operation. Note that the wires shown with dotted lines must be removed from the standard starter. The push button station can be either a Bulletin 800, heavy duty, or a Bulletin 800T, oiltight.

## Plugging



### Plugging a Motor to a Stop from One Direction. — Safety Latch Provided

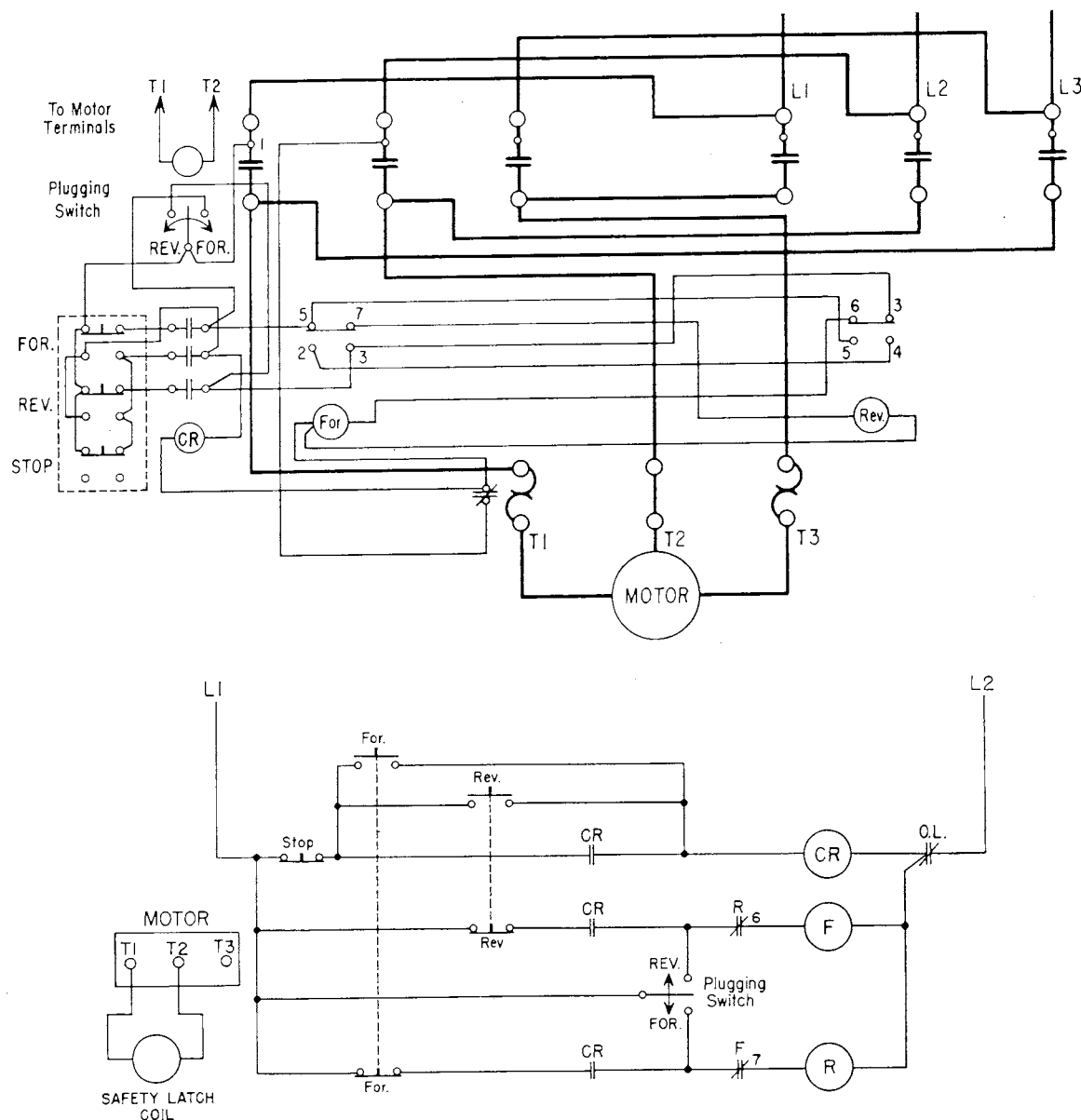
This system is for a motor that is to run in one direction only and must come to an immediate stop when the stop button is pressed. The reverse contactor of the Bulletin 705 reversing switch is used only for plug-stopping and not for running in reverse. When a standard Bulletin 705 is used, wire "W" and all wires represented by dotted lines should be removed.

The safety latch is built into the Bulletin 808 Zero Speed Plugging Switch and its function is to prevent an accidental turn of the motor shaft from closing the plugging switch contacts and starting the motor. This protective feature is optional and the plugging switch can be furnished without safety latch if desired.

The push button station is a Bulletin 800, Type 2HA or a Bulletin 800T, Type 2TA.



### Plugging



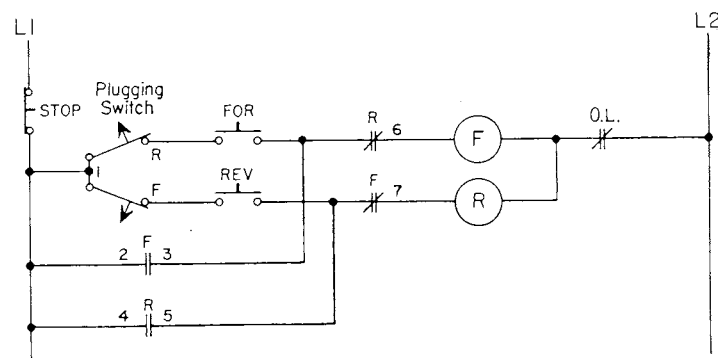
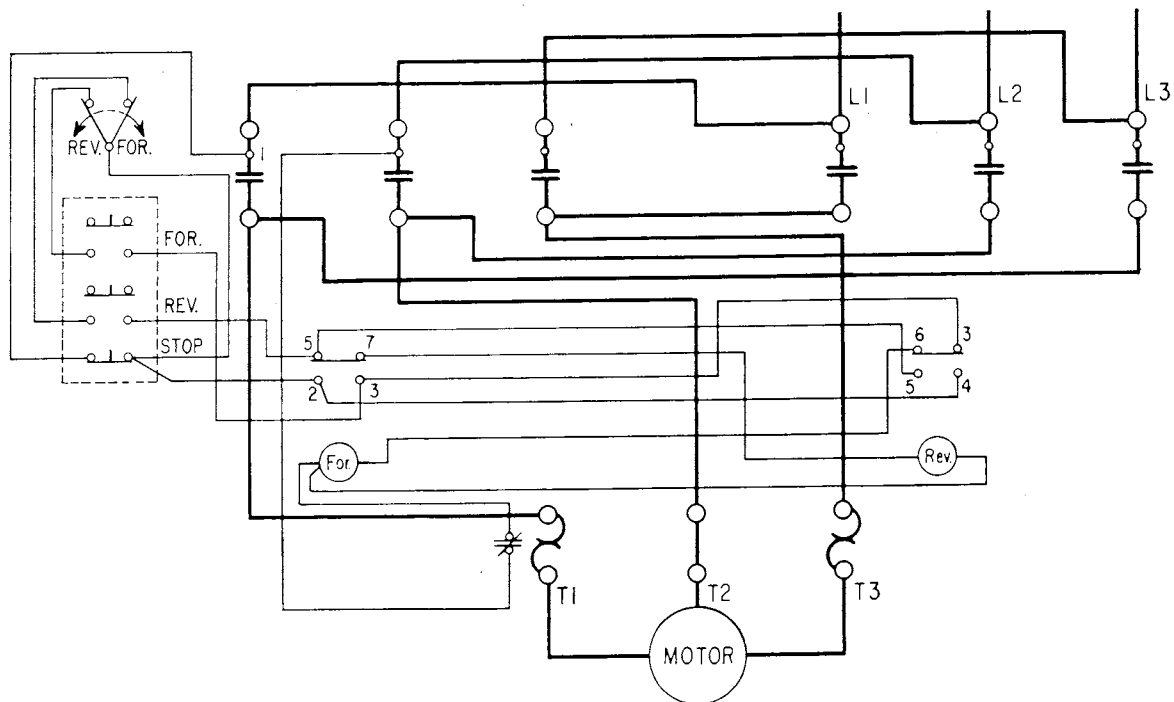
#### Plugging a Motor to a Stop from Either Direction. — Safety Latch Provided

With the system wired as shown, the motor can be started in either direction by pressing the proper button. Pressing the stop button will plug the motor to a stop from either direction. A standard Bulletin 705 reversing switch is used for this application.

The safety latch is a built-in part of the Bulletin 808 Zero Speed Plugging Switch and it prevents an accidental turn of the motor shaft from closing the plugging switch contacts and starting the motor. This protective feature is optional and the plugging switch can be furnished without safety latch if desired.

The control relay necessary with this system is a Bulletin 700, Type B-30 and the push button station can be a Bulletin 800, Type 3HA, or a Bulletin 800T, Type 3TA.

## "Anti-Plugging"



### "Anti-Plugging" — Motor Is to Be Reversed, but it Must Not Be Plugged

A Bulletin 808 Zero Speed Plugging Switch with normally closed contacts is used to prevent plugging. The schematic diagram shows that with the motor operating in one direction, a contact on the Zero Speed Switch opens the control circuit of the starter used for the opposite direction. The open contact will not close until the motor has slowed down, and thus, the reversing switch cannot be energized to change the direction of the motor until the motor is moving slowly.

A standard Bulletin 705 reversing switch can be used with this application. The push button station can be a Bulletin 800, Type 3SA, or 3HA, or a Bulletin 800T, Type 3TA.